

FULL ARTICLE

Momordica charantia reverses type II diabetes in rat

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Abstract

Diabetes, a disease with abnormal production or use of insulin, is a growing concern that affects many individuals globally. Although many attempts have been made, there is no satisfactory treatment for diabetes. Recently, scientists have been exploring a promising treatment of diabetes involving herbal medicine. In this line, we show that *Momordica charantia*, a tendril-bearing vine belonging to the Cucurbitaceae family, permanently normalizes blood glucose levels comparable to healthy rats. Most importantly, *M. charantia* increases the expression of Insulin and Pdx1 genes while lowers the expression Glut2. Moreover, the number and size of the pancreatic islets have remarkably increased in treated animals. Liver ALT, AST, and ALP enzyme activities fell into normal range in treated animals suggesting the protective effect of *M. charantia*. These data indicate that *M. Charantia* improves the pancreas function by activating pancreatic beta cells and protecting liver tissue.

Practical applications

Owing to the effectiveness of *Momordica Charantia* extracts in management of diabetes in STZ-induced diabetic rats, we have intention to evaluate the powder of *Charantia* to discover novel drug for treating diabetes. It is expected that the results could be translated in clinical trials.

KEYWORDS

hepatic Enzymes, momordica, regeneration, type II diabetes

1 | INTRODUCTION

Diabetes mellitus (DM) is a disease with abnormal production or use of insulin characterized by hyperglycemia and deregulated metabolism of biomolecules. Mounting evidence indicates that chronic hyperglycemia leads to diabetes-related microvascular and macrovascular complications such as atherosclerosis, hyperlipidemia, ischemic attacks, retinopathy and nephropathy, which are associated with the increased risk of mortality (Stadler, Jenei, von Bolcschazy, Somogyi, & Jakus, 2003). It has been anticipated that the worldwide incidence of diabetes in adult patients will increase by about 69% from 2010 to 2030 (Mobaseri, Aliasgarzadeh, Bahrami, Zargami, & Tabrizi, 2012; Shaw, Sicree, & Zimmet, 2010). Complete loss and reduced content of β -cells, respectively, in type 1 and type 2 diabetes mellitus (T1DM

and T2DM) are associated with glycosuria, polyuria, polydipsia, slow healing of wounds, and weakness. However, T2DM is a global public health issue particularly in older adults with higher incidence, lower pathogenic severity, and the lack of insulin sensitivity than T1DM (Abegunde, Mathers, Adam, Ortegón, & Strong, 2007; Wild, Roglic, Green, Sicree, & King, 2004). It is a fact that diabetic patients usually consider alternative resources along with common anti-diabetic drugs to control their disease (Romero-Cerecero et al., 2009).

More than 1,200 herbal medicines were identified to have beneficial effect in diabetes (Tarighat Esfanjani, 2012). Several attempts were made to investigate the anti-diabetic mechanism of popular medical herbs and their active ingredients aiming to smooth the way to discover novel anti-diabetic drugs (Gupta et al., 2005; Marles & Farnsworth, 1995). *Momordica charantia* (MC), also known as bitter

gourd or bitter melon, belonging to the Cucurbitaceae family is a tropical plant (Siemonsma & Piluek, 1997). According to several studies, *M. charantia* is able to alter the metabolism of intracellular biomolecules for alleviation of T2DM symptoms (Alam et al., 2015). In this line, the fruits from bitter melon have been demonstrated to exhibit hypoglycemic effects in diabetic animal models (Akhtar, Athar, & Yaqub, 1981; Cakici et al., 1994; Karunanayake, Jeevathayaparan, & Tennekoon, 1990; Rath, Grover, & Vats, 2002; Sarkar, Pranava, & Marita, 1996; Sathishsekar & Subramanian, 2005; Srivastava, 1993; Vikrant, Grover, Tandon, Rath, & Gupta, 2001; Virdi, 2003; Yibchok-anun et al., 2006). It seems that the anti-diabetic mechanisms of bitter melon may be attributed to insulin-like effects (Khanna, Jain, Panagariya, & Dixit, 1981), increase in the secretion of insulin (Welihinda, Arvidson, Gylfe, Hellman, & Karlsson, 1982; Yibchok-anun et al., 2006), regeneration of pancreatic β -cells (Ahmed, Adeghate, Sharma, Pallot, & Singh, 1998), augmentation of glucose uptake in peripheral tissues (Welihinda et al., 1982; Yibchok-anun et al., 2006), reduction in absorption of glucose (Shetty, Kumar, Sambaiah, & Salimath, 2005), rise in glycogen storage from liver and muscle (Welihinda & Karunanayake, 1986; Yibchok-anun et al., 2006), suppression of hepatic gluconeogenesis (Shibib, Khan, & Rahman, 1993), and amelioration in the activity of main enzymes in glycolysis (Rath, 2002).

In the present study, we aimed to investigate hypoglycemic effect of *M. charantia* and its cellular and molecular mechanisms on the pancreas and the liver.

2 | MATERIALS AND METHODS

2.1 | Animals

The rat was used in these studies to investigate the effect of bitter melon extract on pancreas. Experimental procedures were performed in accordance with the Guide to the Care and Use of Experimental Animals and approved by the Animal Care Committee at the Kermanshah Medical University. Animals were randomly divided into three groups: Control group was fed with the diets containing 20% protein, 4.5% fat, 4.5% fiber, 1% calcium, 0.7% phosphorus, 0.5% salt, 0.33% methionine, 0.63 methionine + cysteine, and 0.25% tryptophan. Ripe fruits of *Momordica charantia* were purchased from a medical herb market. The fresh whole fruits were chopped into small pieces and powdered. The experimental group was fed with the diet supplemented with *M. charantia* powder at 10% and 20%.

2.2 | Induction of diabetes

Diabetes was induced in rats after 16 hr of fasting by intra-peritoneal injection of Streptozotocin (Sigma- Aldrich Corp, St. Louis, MO, USA) 35 mg/kg body weight in 0.1 M citrate buffer, pH 4.4. After injection, the rats had free access to food and water. After 72 hr, blood samples were taken via the tail vein before the start of the experiment and repeated every two weeks. The blood glucose (mg/dl) was measured by glucometer (GlucoDr, All Medicus Co, Korea).

Highlight

- *Momordica charantia* normalized blood glucose level in STZ-induced diabetic rats.
- *M. charantia* increased the expression of Insulin, Pdx1 and decreased the expression of Glut2.
- Liver ALT, AST, and ALP enzyme activities fell into normal range in *M. charantia*-treated animals.

2.3 | Histology (Immunocytochemistry)

Animals were euthanized and the pancreases were removed. The pancreatic tissues were preserved in 10% formaldehyde for 4 hr and embedded in paraffin and cut to 4 μ m in thickness and stained for hematoxylin and eosin.

For insulin staining, we used Dako#1633 kit according to the manufacturer's instruction. The sections were examined using light microscope (Nikon BX61-USA) connected to motorized controller unit (Olympus bx-ucb-USA) and photographed by a camera (Nikon DP72-USA).

2.4 | Q-PCR

RNA was extracted using Trizol with classical method (Piotr, 1993). DNA contamination was extracted with an extraction kit (Cat# 35,300) containing DNase. After the measurement of RNA using Nanodrop, cDNA was reverse transcribed by a Superscript III synthesis kit (Takar, Tokyo, Japan). Gene expression assay was carried out using standardized SYBR green in a 7900HT Fast Real-Time PCR System (Applied Biosystems). Comparative quantitation analysis was performed using the delta Ct ($2^{-\Delta Ct}$) method with β -actin and GAPDH as the endogenous controls, and normal pancreas as calibrator. Primers were prepared from Qiagen.

2.5 | Statistical analysis

Statistical analysis was performed using the Statistical Package for Social Sciences (SPSS). Each value is expressed as mean $\pm$ standard deviation (S.D.). One-way analysis of variance (ANOVA) was used to evaluate differences among experimental groups. The results were considered statistically significant if the *p*-values were less than 0.05.

3 | RESULT

3.1 | *M. Charantia* improve the pancreas tissue

Over the course of 6-week treatment, *M. charantia* at 20% concentration significantly decreases the blood sugar levels (average 170 mg/dl) in diabetic rats compared to 10% concentrations, as well as untreated ones (average 500 mg/dl), two-way ANOVA

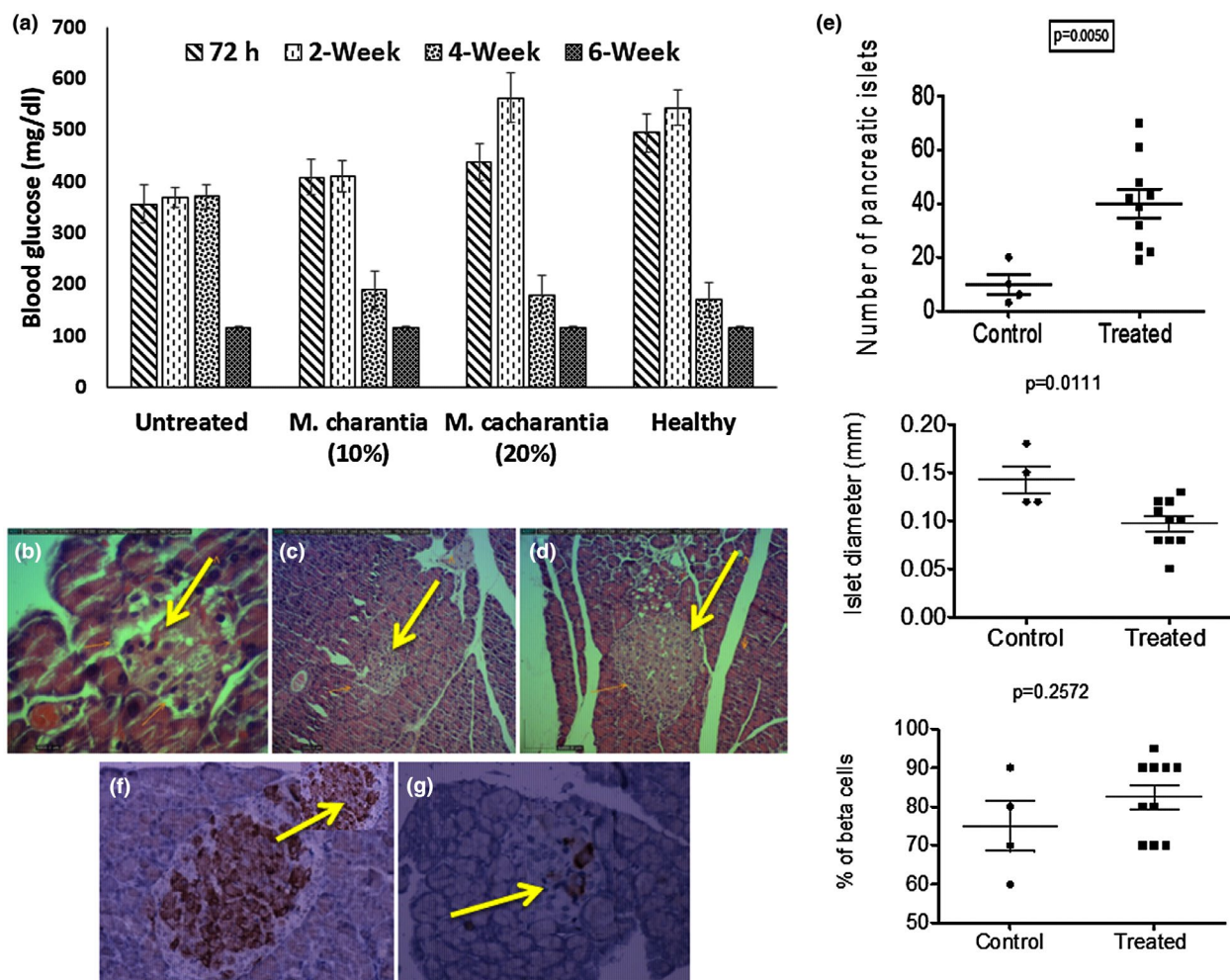


FIGURE 1 *Momordica charantia* lowered the blood sugar levels, while increased insulin expression, as well as the size and number of pancreatic islets. (a) Over the course of 6-week treatment, *Momordica charantia* at 20% concentration significantly decreases the blood sugar levels (average 170 mg/dl) in diabetic rats compared to 10% concentrations, as well as untreated ones (average 500 mg/dl), two-way ANOVA ($p < .0001$). (b–d) Photomicrographs of rat pancreas post 6-week treatment were stained by H&E. (b) Diabetic rats with no treatment as control show small size pancreas (shown by arrow). (c) Diabetic rats with 10% *M. charantia* powder. (d) Diabetic rats with 20% *M. charantia* powder show more and larger size pancreas (yellow arrow). (e) The number of pancreatic islets increased significantly ($p = .005$); however, islets diameter and the percentage of beta cell were not statistically different. (f and g) Photomicrographs of rat pancreas post 6-week

($p < .0001$) (Figure 1a). Photomicrographs of rat pancreas post 6-week treatment were stained by H&E. Diabetic rats with no treatment as control show small size pancreas (shown by arrow). Diabetic rats with 10% *M. charantia* powder are shown in Figure 1b–d. Diabetic rats with 20% *M. charantia* powder show more and larger size pancreas (yellow arrow) (Figure 1e). The number of pancreatic islets increased significantly ($p = .005$); however, islets diameter and the percentage of beta cells were not statistically different (Figure 1 f and g).

Insulin is the most important indicator of the state of diabetes. In order to evaluate the effect of *M. charantia* on the pancreas and the insulin secretion, we measured the insulin secretion and insulin gene expression, and analyzed the histology of pancreas tissue. Animal treated with *M. charantia* remarkably increased insulin secretion (2 IU) compared to untreated diabetic animals (0.6 IU) (Figure 2a). Expression of insulin has significantly increased in

treated animals with 20% *M. charantia* comparable to the healthy rats. Next, we tested the repression genes involved in glucose metabolism. Q-PCR result revealed that insulin, Glucokinase, Pdx1, PTP1b gene expressions were upregulated (Figure 2b). *M. charantia*-treated animals gained body weight. Rats treated with *M. charantia* gained weight, two-way ANOVA ($p = .1769$) (Figure 2c). It is the fact that insulin is involved in regulation of the metabolism of biomolecules through induction the absorption of glucose from the blood into liver, fat, and skeletal muscle cells. Glucokinase as the first enzyme in the glycolysis pathway is able to phosphorylate glucose to trap it in the cells. Protein-tyrosine phosphatase 1B (PTP1B) is an enzyme involved in the regulation of insulin signaling pathway. Finally, we verified the insulin expression with immunocytochemistry and noted more and larger size pancreatic islands in treated rats (Figure 2c and d). Collectively, q-PCR data indicate the reversion of pancreas gland function to normal.

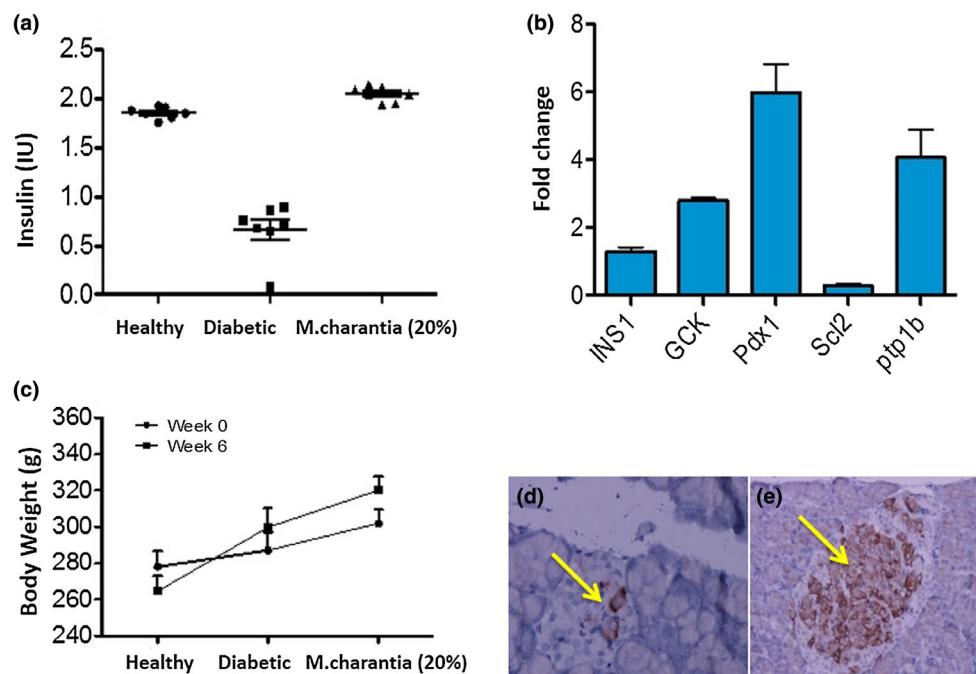


FIGURE 2 *Momordica charantia* restores insulin secretion to normal levels. (a) Animal treated with *Momordica charantia* remarkably increased insulin secretion (2 IU) compared to untreated diabetic animals (0.6 IU), one-way ANOVA ($p < .0001$). (b) Q-PCR showed that insulin and Glucokinase gene expressions were upregulated (the samples were normalized and compared to diabetics, please see methods). (c) *M. charantia*-treated animals gained body weight. Rats treated with *M. charantia* gained weight, two-way ANOVA ($p = .1769$) (d and e) QPCR data were verified by staining: expression of insulin has significantly increased in treated animals with 20% *M. charantia* (yellow arrow)

3.2 | *M. charantia* protects the liver tissue

Liver is among the organs that are affected directly as diabetes progress. In order to explore the function and integrity of the liver, we analyzed the liver enzyme activity and its histology. Interestingly, when animals were exposed to *M. charantia* at 20% concentration, their liver GOT, GPT, and ALP enzyme activities returned to normal levels compared to untreated one ($p = .0022$). No significant changes in enzyme activity observed between 10% and 20% *M. charantia* (Figure 3a). *M. charantia* in doses of 10% (Figure 3c) and 20% (Figure 3d) showed the protective effects at liver of treated diabetic rats compared to untreated diabetic rats (Figure 3b).

4 | DISCUSSION

It was previously shown that *M. Charantia* could lower the blood sugar (Welihinda, Karunanayake, Sheriff, & Jayasinghe, 1986). However, it was not clear whether or not *M. charantia* acts on pancreas directly or acts in a broader stream and targets different organ. Collectively, our data show that *M. charantia* acts on pancreas; upregulates the insulin gene and subsequently insulin secretion. Moreover, it provides a protective effect on the liver, as well.

To our knowledge, this is the first report about *M. charantia* that shed light on the mechanism and the way it delivers its anti-diabetogenic effects in a dose-dependent fashion. This mechanism

essentially represents a novel cellular molecular mechanism for *M. charantia*. While previous studies show contradictory result (Olson & Pessin, 1996), we provide evidence that the *M. charantia* effects are permanent and irreversible. To further investigate the effect of bitter melon, we processed qPCR and looked at some specific genes. Glut1 widely distributed in fetal tissues. While in adult, the high expression of Glut1 is confined to erythrocytes as well as endothelial cells of barrier tissues including the blood-brain barrier. However, the absorption of the low level of basal glucose is the main duty of Glut1 in all cells (Lu et al., 2012). Glut1 expression was decreased in treated animals (Figure 2). The INS gene is necessary for producing insulin to control glucose levels in the blood. INS gene expression has significantly increased in our treated samples (Figure 2b). This explains the lower blood sugar levels in treated rats. Moreover, pancreatic and duodenal homeobox 1 (PDX1) also named insulin promoter factor 1 is a main transcription factor in the development of pancreas to conduct precursor cells into β -cells, and also in duodenal differentiation. As expected, the Pdx1 expression was upregulated dramatically in the treated animals (Figure 2b). Most importantly, the PTP1B enzyme impairs the insulin signaling pathway through dephosphorylation of the insulin receptor and its downstream signal, IRS-1 in the insulin signaling pathway. Therefore, it could be considered as a promising target for treatment of related pathogenesis, particularly type 2 diabetes (Brown-Shimer, 1990). As predicted, PTP1B was upregulated in the treated rats (Figure 3b).

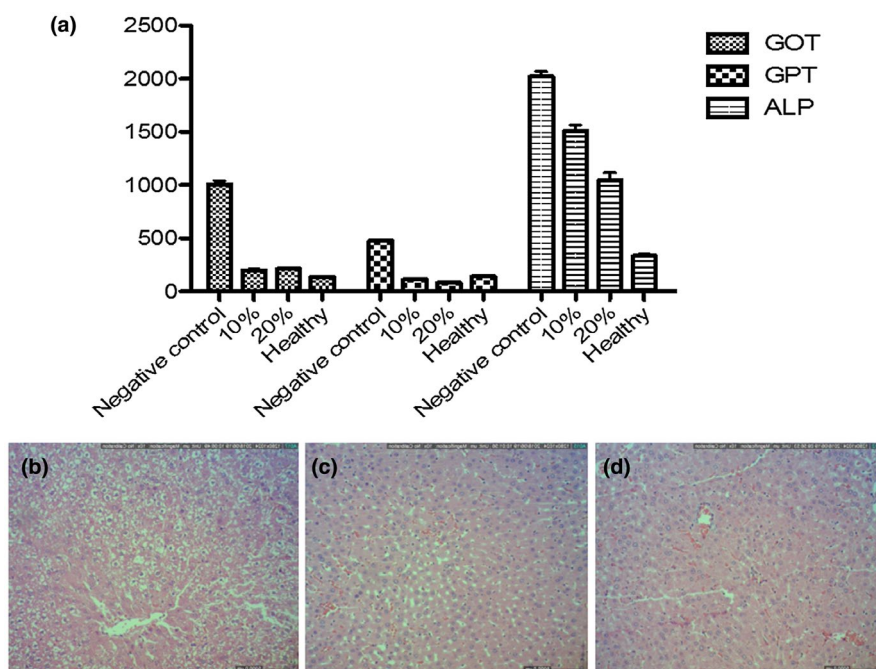


FIGURE 3 *Momordica charantia* protects liver tissue. (a) When rats treated with *M. charantia*, their liver GOT, GPT, and ALP enzyme activities returned to normal levels compared to healthy ones ($p = .0022$). However, ALP levels were increased higher than healthy rats. No significant changes in enzyme activity observed between 10% and 20% *M. charantia*, one-way ANOVA. Photomicrographs of rats' liver stained by Hematoxylin & Eosin: (b) Untreated diabetic rat, (c) diabetic rats with 10% *M. charantia* powder, and (d) diabetic rats with 20% *M. charantia* powder

Dysregulation of lipid metabolism is further supported by the increased risk of non-alcoholic fatty liver disease in patients with both type 1 and 2 diabetes. However, in a reciprocal interplay, fatty liver disease seems to even promote development of type 2 diabetes. Along these lines, in ones with two diseases, poorly managed type 2 diabetes can make fatty liver disease worse. In order to assess the effect of our treatment on liver, we have measured the liver enzymes and assess the inflammation in the liver tissue.

Assessment of hepatic enzymes including AST, ALT and ALP has been suggested to clearly reveal the normal metabolism and function of liver. AST activity was comparable to the healthy animals in the 20% treated rats suggesting a role of bitter melon in protecting liver (Figure 3a). The ALT activity in both treated rats is nearly equal to those in the healthy animals at the end of treatment again, indicating a protective effect of bitter melon on the liver and its function. Our result shows that ALP activity decreases in our treated animals compared to non-intervened animals (Figure 3a). Moreover, quantitative analysis of Ki67 staining in liver revealed the more proliferation in rats treated with 20% *M. Charantia* compared to diabetic rats.

Collectively, our data show that not only does bitter melon restore T2D but also it protects the tissues such as liver from degeneration. It would be informative to further analysis to identify the main ingredient in bitter melon for management of diabetes. Additionally, looking for any effect of bitter melon on trans-differentiation (alpha to beta cell), evaluation of possible effect on the stem cell and differentiation is necessary at the cellular and molecular levels.

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CONFLICT OF INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication.

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REFERENCES

- Abegunde, D. O., Mathers, C. D., Adam, T., Ortegon, M., & Strong, K. (2007). The burden and costs of chronic diseases in low-income and middle-income countries. *Lancet*, 370(9603), 1929–1938. [https://doi.org/10.1016/S0140-6736\(07\)61696-1](https://doi.org/10.1016/S0140-6736(07)61696-1)
- Ahmed, I., Adeghate, E., Sharma, A., Pallot, D., & Singh, J. (1998). Effects of *Momordica charantia* fruit juice on islet morphology in the pancreas of the streptozotocin-diabetic rat. *Diabetes Research and Clinical Practice*, 40(3), 145–151. [https://doi.org/10.1016/S0168-8227\(98\)00022-9](https://doi.org/10.1016/S0168-8227(98)00022-9)

- Akhtar, M. S., Athar, M. A., & Yaqub, M. (1981). Effect of *Momordica charantia* on blood glucose level of normal and alloxan-diabetic rabbits. *Planta Medica*, 42(3), 205–212. <https://doi.org/10.1055/s-2007-971629>
- Alam, M. A., Uddin, R., Subhan, N., Rahman, M. M., Jain, P., & Reza, H. M. (2015). Beneficial role of bitter melon supplementation in obesity and related complications in metabolic syndrome. *Journal of Lipids*, 2015. <https://doi.org/10.1155/2015/496169>
- Brown-Shimer, S., Johnson, K. A., Lawrence, J. B., Johnson, C., Bruskin, A., Green, N. R., & Hill, D. E. (1990). Molecular cloning and chromosome mapping of the human gene encoding protein phosphotyrosyl phosphatase 1B. *Proceedings of the National Academy of Sciences*, 87(13), 5148–5152.
- Cakici, I., Hurmoglu, C., Tunctan, B., Abacioglu, N., Kanzik, I., & Sener, B. (1994). Hypoglycaemic effect of *Momordica charantia* extracts in normoglycaemic or cyproheptadine-induced hyperglycaemic mice. *Journal of Ethnopharmacology*, 44(2), 117–121. [https://doi.org/10.1016/0378-8741\(94\)90077-9](https://doi.org/10.1016/0378-8741(94)90077-9)
- Gupta, R. K., Kesari, A. N., Murthy, P. S., Chandra, R., Tandon, V., & Watal, G. (2005). Hypoglycemic and antidiabetic effect of ethanolic extract of leaves of *Annona squamosa* L. in experimental animals. *Journal of Ethnopharmacology*, 99(1), 75–81. <https://doi.org/10.1016/j.jep.2005.01.048>
- Karunanayake, E. H., Jeevathayaparan, S., & Tennekoon, K. H. (1990). Effect of *Momordica-charantia* fruit juice on streptozotocin-induced diabetes in rats. *Journal of Ethnopharmacology*, 30(2), 199–204. [https://doi.org/10.1016/0378-8741\(90\)90008-H](https://doi.org/10.1016/0378-8741(90)90008-H)
- Khanna, P., Jain, S. C., Panagariya, A., & Dixit, V. P. (1981). Hypoglycemic activity of polypeptide-p from a plant source. *Journal of Natural Products*, 44(6), 648–655.
- Lu, L., Seidel, C. P., Iwase, T., Stevens, R. K., Gong, Y. Y., Wang, X., ... Campochiaro, P. A. (2012). Suppression of GLUT1; A new strategy to prevent diabetic complications. *Cellular Physiology*, 228(2), 251–257.
- Marles, R. J., & Farnsworth, N. R. (1995). Antidiabetic plants and their active constituents. *Phytomedicine*, 2(2), 137–189. [https://doi.org/10.1016/S0944-7113\(11\)80059-0](https://doi.org/10.1016/S0944-7113(11)80059-0)
- Mobaseri, M., Aliasgarzadeh, A., Bahrami, A., Zargami, N., & Tabrizi, A. (2012). Efficacy of the total extract of *Urtica dioica* on the glucose utilization by the human muscle cells. *Journal of Clinical and Diagnostic Research*, 6(3), 437–440.
- Olson, A. L., & Pessin, J. E. (1996). Structure, function, and regulation of the mammalian facilitative glucose transporter gene family. *Annual Review of Nutrition*, 16, 235. <https://doi.org/10.1146/annurev.nu.16.070196.001315>
- Piotr, C. (1993). A reagent for the single-step simultaneous isolation of RNA, DNA and proteins from cell and tissue samples. *BioTechniques*, 15(3), 532.
- Rathi, S. S., Grover, J. K., & Vats, V. (2002). The effect of *Momordica charantia* and *Mucuna pruriens* in experimental diabetes and their effect on key metabolic enzymes involved in carbohydrate metabolism. *Phytotherapy Research*, 16, 7. <https://doi.org/10.1002/ptr.842>
- Romero-Cerecero, O., Reyes-Morales, H., AguilarSantamaria, L., Huerta-Reyes, M., Tortoriello-Garcia, J., & Hortensia, R.-M. (2009). Use of medicinal plants among patients with diabetes mellitus type 2 in Morelos, Mexico. *Use of Medicinal Plants in Morelos, Mexico*, 8(5), 8.
- Sarkar, S., Pranava, M., & Marita, R. (1996). Demonstration of the hypoglycemic action of *Momordica charantia* in a validated animal model of diabetes. *Pharmacological Research*, 33(1), 1–4. <https://doi.org/10.1006/phrs.1996.0001>
- Sathishsekar, D., & Subramanian, S. (2005). Beneficial effects of *Momordica charantia* seeds in the treatment of STZ-induced diabetes in experimental rats. *Biological & Pharmaceutical Bulletin*, 28(6), 978–983. <https://doi.org/10.1248/bpb.28.978>
- Shaw, J. E., Sicree, R. A., & Zimmet, P. Z. (2010). Global estimates of the prevalence of diabetes for 2010 and 2030. *Diabetes Research and Clinical Practice*, 87(1), 4–14. <https://doi.org/10.1016/j.diabetes.2009.10.007>
- Shetty, A. K., Kumar, G. S., Sambaiha, K., & Salimath, P. V. (2005). Effect of bitter gourd (*Momordica charantia*) on glycaemic status in streptozotocin induced diabetic rats. *Plant Foods for Human Nutrition*, 60(3), 109–112. <https://doi.org/10.1007/s11130-005-6837-x>
- Shibib, B. A., Khan, L. A., & Rahman, R. (1993). Hypoglycaemic activity of *Coccinia indica* and *Momordica charantia* in diabetic rats: depression of the hepatic gluconeogenic enzymes glucose-6-phosphatase and fructose-1, 6-bisphosphatase and elevation of both liver and red-cell shunt enzyme glucose-6-phosphate dehydrogenase. *Biochemical Journal*, 292(1), 267–270.
- Siemonsma, J., & Piluek, K. (1997). *Plants resources of South-East Asia [electronic resource]*. World biodiversity database CD-ROM series: UNESCO-Publishing.
- Srivastava, Y., Venkatakrishna-Bhatt, H., Verma, Y., Venkaiah, K., & Raval, B. H. (1993). Antidiabetic and adaptogenic properties of *Momordica Charantia* extract: An experimental and clinical evaluation. *Phytotherapy Research*, 7, 4. <https://doi.org/10.1002/ptr.2650070405>
- Stadler, K., Jenei, V., von Bolcschazy, G., Somogyi, A., & Jakus, J. (2003). Increased nitric oxide levels as an early sign of premature aging in diabetes. *Free Radical Biology and Medicine*, 35(10), 1240–1251. [https://doi.org/10.1016/S0891-5849\(03\)00499-4](https://doi.org/10.1016/S0891-5849(03)00499-4)
- Tarighat Esfajani, A., Namazi, N., Bahrami, A., & Ehteshami, M. (2012). Effect of hydroalcoholic extract of nettle (*Urtica dioica*) on glycemic index and insulin resistance index in type 2 diabetic patients. *Iranian Journal of Endocrinology and Metabolism*, 6(13), 7.
- Vikrant, V., Grover, J. K., Tandon, N., Rathi, S. S., & Gupta, N. (2001). Treatment with extracts of *Momordica charantia* and *Eugenia jambolana* prevents hyperglycemia and hyperinsulinemia in fructose fed rats. *Journal of Ethnopharmacology*, 76(2), 139–143. [https://doi.org/10.1016/S0378-8741\(01\)00218-5](https://doi.org/10.1016/S0378-8741(01)00218-5)
- Virdi, J., Sivakami, S., Shahani, S., Suthar, A. C., Banavalikar, M. M., & Biyani, M. K. (2003). Antihyperglycemic effects of three extracts from *Momordica charantia*. *Journal of Ethnopharmacology*, 88, 4. [https://doi.org/10.1016/S0378-8741\(03\)00184-3](https://doi.org/10.1016/S0378-8741(03)00184-3)
- Welihinda, J., Arvidson, G., Gylfe, E., Hellman, B., & Karlsson, E. (1982). The insulinreleasing activity of the tropical plant *Momordica charantia*. *Acta Biologica Et Medica Germanica*, 41, 11.
- Welihinda, J., & Karunanayake, E. H. (1986). Extra-pancreatic effects of *Momordica charantia* in rats. *Journal of Ethnopharmacology*, 17(3), 247–255. [https://doi.org/10.1016/0378-8741\(86\)90112-1](https://doi.org/10.1016/0378-8741(86)90112-1)
- Welihinda, J., Karunanayake, E. H., Sheriff, M. H., & Jayasinghe, K. S. (1986). Effect of *Momordica charantia* on the glucose tolerance in maturity onset diabetes. *Journal of Ethnopharmacology*, 17(3), 277–282. [https://doi.org/10.1016/0378-8741\(86\)90116-9](https://doi.org/10.1016/0378-8741(86)90116-9)
- Wild, S., Roglic, G., Green, A., Sicree, R., & King, H. (2004). Global prevalence of diabetes: Estimates for the year 2000 and projections for 2030. *Diabetes Care*, 27(5), 1047–1053. <https://doi.org/10.2337/diacare.27.5.1047>
- Yibchok-anun, S., Adisakwattana, S., Yao, C. Y., Sangvanich, P., Roengsumran, S., & Hsu, W. H. (2006). Slow acting protein extract from fruit pulp of *Momordica charantia* with insulin secretagogue and insulinomimetic activities. *Biological & Pharmaceutical Bulletin*, 29(6), 1126–1131. <https://doi.org/10.1248/bpb.29.1126>

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